

MATH3018 Assignment 8

- (1) [4pts] Consider two sample sets X_1 and X_2 from two bivariate normal populations C_1 and C_2 , respectively:

$$X_1 = \begin{pmatrix} 3 & 7 \\ 2 & 4 \\ 4 & 7 \end{pmatrix}, X_2 = \begin{pmatrix} 6 & 9 \\ 5 & 7 \\ 4 & 8 \end{pmatrix}.$$

Assume a same population covariance matrix for C_1 and C_2 . Classify the observation $x_0 = (2, 7)^T$.

[Solution]

By calculations $\bar{x}_1 = (3, 6)^T$, $\bar{x}_2 = (5, 8)^T$, and

$$S_{\text{pooled}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

$$\begin{aligned} \delta_{12}(x) &= (\bar{x}_1 - \bar{x}_2)^T S_{\text{pooled}}^{-1} x - \frac{1}{2} (\bar{x}_1 - \bar{x}_2)^T S_{\text{pooled}}^{-1} (\bar{x}_1 + \bar{x}_2) + \log(\hat{\pi}_2/\hat{\pi}_1) \\ &= (-2, -2) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \frac{1}{2} (-2, -2) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 14 \end{pmatrix} + \log(0.5/0.5) \\ &= -2x_1 + 8 \end{aligned}$$

Since $\delta_{12}(x_0) = 4 > 0$, we classify x_0 into C_1 .

- (2) [4pts] Let $f_1(x) = P(X = x|Y = 1) = \frac{1}{2}(1 - |x|)$ for $|x| \leq 1$ and $f_2(x) = P(X = x|Y = 2) = \frac{1}{2}(1 - |x - .5|)$ for $-.5 \leq x \leq 1.5$.

(a) Sketch the two densities.

(b) Identify the classification regions of Bayes classifier when the prior probability of class $\pi_1 = \pi_2$.

(c) Identify the classification regions when $\pi_1 = .2$

[Solution]

(a) Omitted.

(b) Given $\pi_1 = \pi_2$, we have $\pi_2 f_2(x) \leq \pi_1 f_1(x)$ is equivalent to $f_1(x)/f_2(x) \geq 1$. Therefore $R_1 = [-1, 0.25]$ and $R_2 = (0.25, 1.5]$.

(c) Given $\pi_1 = 0.2$ thus $\pi_2 = 0.8$, we have $\pi_2 f_2(x) \leq \pi_1 f_1(x)$ is equivalent to $f_1(x)/f_2(x) \geq 4$. Therefore $R_1 = [-1, -1/3]$ and $R_2 = (-1/3, 1.5]$.

- (3) [4pts] Consider random samples from three bivariate normal populations C_1, C_2 , and C_3 with the same population covariance matrices. The sample mean vectors and covariance matrices, are as follows:

$$C_1 : n_1 = 3, \quad \bar{\mathbf{x}}_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad \mathbf{S}_1 = \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$$

$$C_2 : n_2 = 3, \quad \bar{\mathbf{x}}_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$$

$$C_3 : n_3 = 3, \quad \bar{\mathbf{x}}_3 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad \mathbf{S}_3 = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$$

Given that $\pi_1 = \pi_2 = 0.25$, $\pi_3 = 0.5$, classify the observation $x_0 = (2, -1)^\top$ using QDA.

[Solution] See Example 11.10 in the reference book: Applied Multivariate Statistical Analysis (6 edition) by Richard Johnson and Dean Wichern.

- (4) [3pts] Suppose that we have prior probability for two classes C_1 and C_2 given as π_1 and π_2 , and the density of $\mathbf{x}$ in C_1 and C_2 are given as $f_1(x)$ and $f_2(x)$, respectively. Show that by classifying those $\mathbf{x}$ satisfying $\pi_1 f_1(\mathbf{x}) \geq \pi_2 f_2(\mathbf{x})$ as C_1 and $\pi_1 f_1(\mathbf{x}) < \pi_2 f_2(\mathbf{x})$ as C_2 achieves the minimum value of TPM (total probability of misclassification).

[Solution] Let $\Omega = \mathbb{R}^p$ be the sample space—that is, the collection of all possible observations $\mathbf{x}$. Let R_1 be that set of $\mathbf{x}$ values for which we classify objects as C_1 and $R_2 = \Omega - R_1$ be the remaining $\mathbf{x}$ values for which we classify objects as C_2 . We know that

$$P(2 | 1) = P(\mathbf{X} \in R_2 | \text{true class } C_1) = \int_{R_2 = \Omega - R_1} f_1(\mathbf{x}) d\mathbf{x},$$

and similarly

$$P(1 | 2) = P(\mathbf{X} \in R_1 | \text{true class } C_2) = \int_{R_1 = \Omega - R_2} f_2(\mathbf{x}) d\mathbf{x}$$

Substituting the integral expressions for $P(2 | 1)$ and $P(1 | 2)$ into $TPM = \pi_1 P(2 | 1) + \pi_2 P(1 | 2)$ yields

$$TPM = \pi_1 \int_{R_2} f_1(\mathbf{x}) d\mathbf{x} + \pi_2 \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$

Noting that $\Omega = R_1 \cup R_2$, so that the total probability

$$1 = \int_{\Omega} f_1(\mathbf{x}) d\mathbf{x} = \int_{R_1} f_1(\mathbf{x}) d\mathbf{x} + \int_{R_2} f_1(\mathbf{x}) d\mathbf{x}$$

we can write

$$TPM = \pi_1 \left[1 - \int_{R_1} f_1(\mathbf{x}) d\mathbf{x} \right] + \pi_2 \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$

By the additive property of integrals (volumes),

$$TPM = \int_{R_1} [\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})] d\mathbf{x} + \pi_1$$

Now, π_1, π_2 are nonnegative. In addition, $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are nonnegative for all $\mathbf{x}$ and are the only quantities in TPM that depend on $\mathbf{x}$. Thus, TPM is minimized if R_1 includes those values $\mathbf{x}$ for which the integrand

$$[\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})] \leq 0$$

and excludes those $\mathbf{x}$ for which this quantity is positive.